CHM-DSM-401 — Fundamental of Chemistry-IV

Unit I: Data Analysis

7th Semester, Assam University (FYUG) — Haflong Government College

Paper: CHM-DSM-401, Full Marks 100 [ESE 70 + Internal 30] | Pass marks 40 | Unit I: 9 lectures

Notes 16 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

Index

Unit syllabus (as in the syllabus file)

1. Sampling, evaluation of analytical data.
2. Errors in chemical analysis, precision and accuracy.
3. Mean and standard deviation, systematic and random errors.
4. Linear regression, covariance and correlation coefficient.
5. Significant figures.

Top 2 high-mark topics in this unit

1. Errors in chemical analysis - systematic and random errors, precision vs accuracy, propagation of errors — core definitions plus numericals, and AU Analytic Methods asks this in Section A and Section B.
2. Mean and standard deviation, linear regression, covariance and correlation coefficient — the only numerical/statistical long-answer block in the unit, and AU asks normal distribution and statistical tests of data analysis as a 5-mark question.

Contents

1. Sampling and Evaluation of Analytical Data
2. Errors in Chemical Analysis: Precision and Accuracy
3. Systematic and Random Errors
4. Mean and Standard Deviation
5. Linear Regression, Covariance and Correlation Coefficient
6. Significant Figures
7. Questions — Short form, then Long form (8 AU questions, from CHMDSE-501T)
8. Sources

1. Sampling and Evaluation of Analytical Data

Sampling in Chemical Analysis

Sampling is the process of collecting a small, manageable portion of material from a larger bulk mass in such a way that the collected portion accurately represents the composition of the entire bulk lot. In quantitative chemical analysis, the final analytical measurement is performed on a very small fraction of the original material. Consequently, the valid analytical result of a sample is only as accurate as the sampling procedure used to obtain it.

The fundamental requirement of any sampling process is representativeness. A sample must represent the whole bulk material because the analytical conclusions drawn from the laboratory measurement are extrapolated directly to the entire batch. If a sample is non-representative, even the most precise and accurate laboratory analysis produces an incorrect conclusion about the target material. Representativeness ensures that the concentration of every chemical component in the sample equals the average concentration of that component in the parent bulk material within acceptable statistical limits.

Different analytical scenarios require specific sampling strategies depending on the homogeneity and spatial distribution of the bulk material:

* **Random Sampling:** In random sampling, individual units or portions are selected from the bulk population in such a manner that every portion has an equal, non-zero probability of being chosen. This approach eliminates operator bias and is ideal for homogeneous or well-mixed populations. * **Systematic Sampling:** In systematic sampling, samples are collected at predetermined, regular intervals of space or time. Examples include collecting a portion of a liquid effluent every two hours or selecting every hundredth manufactured tablet from a continuous conveyor line. * **Stratified Sampling:** When a bulk lot is clearly non-uniform and consists of distinct, identifiable sub-populations or zones (strata) of differing composition, the bulk is partitioned into these individual strata. Random samples are then collected independently from within each stratum in proportion to the relative mass or volume of that stratum within the whole population. * **Composite Sampling:** Composite sampling involves collecting multiple primary samples (grab samples) from different spatial points or at different time intervals across the bulk lot, and thoroughly blending them together to form a single pooled sample. This single sample provides an average analytical value for the entire bulk, reducing total laboratory workload.

The physical state of the bulk material determines the sampling methodology:

* **Sampling of Solids:** Bulk solids are often heterogeneous and prone to segregation according to particle size and density. Sampling particulate solids requires collecting numerous small portions from all parts of the bulk. The combined material is then crushed, ground, passed through sieves, and blended. Homogenized particulate solids are often reduced in mass using the method of coning and quartering to obtain a representative laboratory sample. * **Sampling of Liquids:** Homogeneous single-phase liquids require minimal sampling effort, as a simple grab sample is representative. Heterogeneous liquids, suspension mixtures, or immiscible liquid layers require vigorous mixing prior to sampling. Alternatively, specialized sampling devices such as weighted bottles or thief samplers are lowered to specified depths to collect samples from discrete liquid strata. * **Sampling of Gases:** Gas sampling involves collecting representative vapor phases while accounting for variations in temperature,

pressure, and condensation. Gases are collected using evacuated expansion bulbs, gas-tight syringes, or by drawing the gas stream through liquid absorption traps, cryogenic traps, or solid sorbent tubes that trap specific gaseous analytes for subsequent analysis.

Errors introduced by poor sampling are often the largest source of uncertainty in an overall chemical measurement. Sampling errors arise when the sample fails to reflect the bulk composition due to particle segregation, local concentration gradients, or contamination during sample collection and handling. Furthermore, sample degradation during storage—such as oxidation, absorption of atmospheric moisture, loss of volatile components, or reaction with container walls— permanently alters the chemical composition of the sample. Unlike laboratory measurement errors, sampling errors cannot be corrected by recalibrating laboratory instruments; they can only be minimized through rigorous sampling design.

Evaluation of Analytical Data

Judging the quality of analytical results requires evaluating experimental data using statistical tools. Because repeated measurements of the same chemical sample under identical conditions yield slightly different values due to random variations, statistical distributions are applied to interpret the data.

The Normal Distribution of Random Errors

When an analytical measurement is repeated a large number of times, small, unavoidable random errors accumulate independently. These random variations cause the measured values to distribute symmetrically around the true population mean, forming a bell-shaped curve known as the normal distribution or Gaussian distribution.

The continuous probability density function $f(x)$ of a normal distribution is defined by the population mean μ and the population standard deviation σ:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

In this mathematical model, the population mean μ represents the central location of the distribution, while the population standard deviation σ measures the width or spread of the curve.

The properties of the normal distribution are summarized by the Empirical Rule, commonly known as the 68–95–99.7 rule: * Approximately 68.3% of all measured data points fall within one standard deviation of the mean, represented by the interval $[\mu - 1\sigma, \mu + 1\sigma]$. * Approximately 95.4% of all measured data points fall within two standard deviations of the mean, represented by the interval $[\mu - 2\sigma, \mu + 2\sigma]$. * Approximately 99.7% of all measured data points fall within three standard deviations of the mean, represented by the interval $[\mu - 3\sigma, \mu + 3\sigma]$.

Consider repeated titration readings as a practical example. A student performs a series of replicate titrations to determine the endpoint volume of a standardized sodium hydroxide solution required to neutralize an acid. Small, random variations—such as estimating the last decimal place on the burette scale, minor temperature fluctuations altering solution volume, or slight differences in visual endpoint color recognition—cause individual volume readings to scatter about the mean volume. When a large set of titration readings is plotted as a frequency histogram, it forms a normal distribution centered at the sample mean volume.

Statistical Tests in Analytical Data Analysis

Statistical significance tests allow analysts to make objective decisions about data quality, out-of-spec values, and method performance.

The Q-Test (Dixon's Q-Test): The purpose of the Q-test is to evaluate a single outlying observation (an extraordinarily high or low data value in a small replicate dataset) to determine whether it should be statistically rejected or retained.

To calculate the experimental test statistic Q_{calc}, the data points are arranged in increasing numerical order. The gap is defined as the absolute difference between the suspect outlier x_{suspect} and its nearest neighboring value x_{nearest}:

$$\text{Gap} = |x_{\text{suspect}} - x_{\text{nearest}}|$$

The range of the entire dataset is the difference between the maximum value x_{max} and the minimum value x_{min}:

$$\text{Range} = x_{\text{max}} - x_{\text{min}}$$

Dividing the absolute gap by the range gives the expression for Q_{calc}:

$$Q_{\text{calc}} = \frac{|x_{\text{suspect}} - x_{\text{nearest}}|}{x_{\text{max}} - x_{\text{min}}}$$

The calculated value Q_{calc} is compared with a critical value Q_{critical} obtained from statistical tables for a given number of observations n and a chosen confidence level (typically 95%). If $Q_{\text{calc}} > Q_{\text{critical}}$, the suspect data point is rejected as an outlier; otherwise, it must be retained.

The t-Test (Student's t-Test): The purpose of the t-test is to compare an experimentally determined sample mean $\bar{x}$ against a known, accepted reference value μ_0, or to compare the sample means of two independent sets of measurements ($\bar{x}_1$ and $\bar{x}_2$) to determine if there is a statistically significant difference between them.

For comparing a single sample mean $\bar{x}$ derived from n replicate measurements having a sample standard deviation s against a true reference value μ_0, the calculated statistic t_{calc} is defined as:

$$t_{\text{calc}} = \frac{|\bar{x} - \mu_0|\sqrt{n}}{s}$$

If the calculated t_{calc} value exceeds the critical table value t_{critical} at the specified degrees of freedom ($n - 1$) and chosen confidence level, the difference between the experimental mean and the reference value is statistically significant, indicating the presence of systematic error.

The F-Test: The purpose of the F-test is to compare the precision of two distinct sets of analytical data, or two analytical methods, by evaluating their sample variances.

The sample variance is defined as the square of the sample standard deviation s^2. The experimental statistic F_{calc} is constructed as the ratio of the larger sample variance to the smaller sample variance:

$$F_{\text{calc}} = \frac{s_1^2}{s_2^2}$$

where the sample standard deviations are ordered such that $s_1^2 \geq s_2^2$. The calculated F_{calc} value is compared with the critical statistic F_{critical} from F-distribution tables for the corresponding degrees of freedom ($n_1 - 1$ and $n_2 - 1$). If $F_{\text{calc}} > F_{\text{critical}}$, the hypothesis of equal precision is rejected, showing that one analytical method is significantly more precise than the other.

2. Errors in Chemical Analysis: Precision and Accuracy

Definition and Expressions of Error

In chemical analysis, an error is defined as the quantitative difference between an experimentally measured value x_i (or an experimental sample mean $\bar{x}$) and the true or accepted reference value x_t.

$$\text{Error} = x_i - x_t$$

Error indicates the degree of inaccuracy present in an experimental measurement. Error is quantitatively expressed in several ways depending on whether absolute or relative comparisons are required.

Absolute Error

The absolute error E_a is the direct numerical difference between the experimental value x_i and the true value x_t. It retains the original units of the physical measurement:

$$E_a = x_i - x_t$$

If an experimental sample mean $\bar{x}$ is evaluated instead of a single measurement, the absolute error of the mean is written as:

$$E_a = \bar{x} - x_t$$

Relative Error

The relative error E_r is a dimensionless ratio that compares the absolute error directly to the magnitude of the true value x_t. The expression for absolute error E_a is substituted into the numerator:

$$E_r = \frac{E_a}{x_t}$$

Substituting $E_a = x_i - x_t$ into the ratio yields:

$$E_r = \frac{x_i - x_t}{x_t}$$

Ways of Expressing Error

Depending on scale and convention, error is expressed in the following quantitative forms:

* **Absolute Error (E_a):** Expressed directly in measurement units:

$$E_a = x_i - x_t$$

* **Relative Error (E_r):** Expressed as a unitless decimal fraction:

$$E_r = \frac{x_i - x_t}{x_t}$$

* **Percentage Error ($\%$ Error):** Expresses the relative error per hundred units by multiplying the relative error fraction by 100:

$$\% \text{ Error} = \left(\frac{x_i - x_t}{x_t} \right) \times 100$$

* **Parts Per Thousand Error (ppt Error):** Expresses the relative error per thousand units by multiplying the relative error fraction by 1000, which is useful for small error values:

$$\text{ppt Error} = \left(\frac{x_i - x_t}{x_t} \right) \times 1000$$

Precision and Accuracy

Precision and accuracy describe two distinct characteristics of analytical data.

Precision: Precision defines the agreement or closeness among multiple replicate measurements that have been obtained in exactly the same way using identical experimental procedures. Precision reflects the reproducibility of an analytical method. High precision means that repeated trials yield values that are closely clustered together. Precision is expressed quantitatively through parameters such as range, sample standard deviation, sample variance, or coefficient of variation. Precision does not depend on the true value x_t.

Accuracy: Accuracy defines the closeness of an experimentally measured value x_i or a sample mean $\bar{x}$ to the accepted true value x_t. Accuracy measures the correctness of a result. It is expressed quantitatively using absolute error or relative error. High accuracy means that the measured result is very close to the true value.

Target-and-Arrow Analogy

The fundamental distinction between precision and accuracy is illustrated using the target-and-arrow example, where the center bullseye of a target represents the true value x_t and each shot arrow represents an individual experimental measurement x_i:

* **High Precision and High Accuracy:** All arrows land in a tight cluster directly inside the central bullseye. The individual shots are reproducible (high precision) and center on the true target value (high accuracy). * **High Precision and Low Accuracy:** All arrows land in a tight cluster, but the cluster is positioned far to one side of the bullseye. The shots are reproducible (high precision), but they systematically miss the true target value (low accuracy), indicating the presence of a systematic error. * **Low Precision and High Accuracy:** The arrows are scattered widely over the target surface, but their spatial center of gravity (the average position of all arrows) is located precisely over the bullseye. Individual measurements lack reproducibility (low precision), but their mathematical mean is close to the true value (high accuracy). * **Low Precision and Low Accuracy:** The arrows

are scattered randomly across the target with no tight clustering and a center of gravity far from the bullseye. The measurements are unrepeatable (low precision) and incorrect (low accuracy).

Minimization of Errors

Experimental errors in chemical analysis cannot be entirely eliminated, but systematic and random errors can be minimized using standard laboratory procedures:

* **Calibration of Instruments:** Analytical instruments such as electronic balances, pH meters, spectrophotometers, and volumetric glassware must be periodically calibrated using reliable standards. Calibration eliminates systematic instrumental errors by establishing a correct functional relationship between the instrument response and the physical value being measured. * **Running Blanks:** A blank determination involves carrying out every step of the analytical procedure without adding the sample itself. The blank contains all reagents, solvents, and indicators exposed to the same glassware and environmental conditions. Subtracting the resulting blank signal from the total analytical sample signal eliminates constant systematic errors caused by impurities in reagents or background signal contributions. * **Replicate Measurements:** Performing replicate analyses (at least three to five independent trials of the same sample) minimizes the effect of random (indeterminate) errors. Because random errors follow a normal distribution, taking the mean $\bar{x}$ of multiple replicates provides an estimate of the true value that is more reliable than any single measurement. The standard error of the sample mean decreases proportionally to $1/\sqrt{n}$, where n is the number of replicate measurements. * **Using Standard Reference Materials (SRMs):** Analyzing certified Standard Reference Materials—substances containing precisely known, matrix-matched analyte concentrations certified by national standardizing bodies—under the exact experimental conditions used for unknown samples allows analysts to detect hidden systematic errors. If the analysis of an SRM yields a result that deviates from its certified value, a correction factor or procedural adjustment is applied. * **Careful Technique:** Strict adherence to established laboratory procedures eliminates personal and operational errors. Careful technique includes allowing heated containers to cool in a desiccator before weighing, avoiding parallax errors when reading volumetric meniscus levels, preventing sample contamination, and ensuring complete chemical reactions.

3. Systematic and Random Errors

Experimental measurements in quantitative chemical analysis are invariably accompanied by errors. These errors determine the reliability, accuracy, and precision of analytical results. Errors in chemical analysis are broadly classified into systematic (determinate) errors and random (indeterminate) errors.

Systematic (Determinate) Errors

Systematic errors, also termed determinate errors, are non-random errors that possess a definite cause and affect all measurements in a replicate series in a similar manner. A key feature of systematic errors is that they shift all measured values in one direction relative to the true value, causing a persistent positive or negative bias. Because systematic errors are unidirectional and reproducible under identical conditions, they cannot be eliminated or reduced simply by repeating the measurement.

Systematic errors are generally categorized based on their variation with sample size into constant errors and proportional errors:

1. **Constant Errors:** The magnitude of a constant error remains fixed regardless of the amount of sample taken for analysis. As the sample size or analyte amount increases, the absolute error stays constant, which means the relative error decreases proportionally.
2. **Proportional Errors:** The absolute magnitude of a proportional error varies directly with the quantity of analyte or sample mass analyzed. Consequently, the relative error remains constant irrespective of the sample size.

The primary sources of systematic errors in chemical analysis include:

- **Instrumental Errors:** Caused by imperfections, lack of calibration, or instability in measuring devices such as uncalibrated glassware, uncalibrated electronic balances, or drifting spectrophotometer detectors. - **Method Errors:** Originate from non-ideal chemical or physical behavior of analytical systems, such as incomplete chemical reactions, side reactions of reagents, coprecipitation of impurities, or thermal instability of precipitates during gravimetric analysis. - **Personal Errors:** Arise from individual human limitations, bad habits, or subjective biases, such as consistently reading a volumetric burette from an incorrect angle (parallax error) or misjudging the visual endpoint color change during a titration. - **Reagent Errors:** Introduced by impurities present in the chemical reagents or solvents used, or by unwanted reactions occurring between reagents and storage containers.

An example of a systematic error is using a volumetric pipette calibrated at $20^\circ\mathrm{C}$ to measure solution volumes at $35^\circ\mathrm{C}$ without temperature correction. Thermal expansion of the liquid causes the delivered volume to be consistently larger than the nominal volume, introducing a constant positive error in every volumetric trial.

Random (Indeterminate) Errors

Random errors, also known as indeterminate errors, arise from uncontrolled, unpredictable, and unavoidable small fluctuations in environmental or physical variables during measurement. Unlike systematic errors, random errors cause measured values to scatter symmetrically on both sides of the mean value.

Random errors follow a Gaussian or normal distribution curve in the limit of a large number of replicate measurements. Because small positive and negative random deviations occur with equal probability, random errors cannot be eliminated individually. However, their effect on the reported result can be minimized by replicating measurements and taking the arithmetic mean.

The sources of random errors include minor variations in ambient room temperature, slight mechanical vibrations in analytical balances, electric noise in instrument circuits, and minor subjective variations in aligning liquid meniscus marks with graduation lines.

An example of a random error is the minor fluctuation in consecutive balance mass readings of a desiccated crucible caused by invisible convective air currents inside the balance chamber.

Propagation of Errors

When individual analytical measurements containing uncertainties are combined mathematically to derive a final calculated result, the error propagates through the calculation. The manner in which individual errors propagate depends on the mathematical operations involved.

Addition and Subtraction

When measured quantities containing random absolute errors are added or subtracted, the absolute standard deviation of the final result is obtained by combining the absolute errors of individual terms in quadrature (the square root of the sum of the squares of individual absolute errors).

Consider a calculated quantity y derived from the addition and subtraction of measured quantities x_1, x_2, and x_3:

$$y = x_1 + x_2 - x_3$$

Let s_{x_1}, s_{x_2}, and s_{x_3} represent the absolute standard deviations of x_1, x_2, and x_3, respectively. The absolute standard deviation of y, denoted s_y, is expressed by:

$$s_y = \sqrt{s_{x_1}^2 + s_{x_2}^2 + s_{x_3}^2}$$

To demonstrate error propagation in addition and subtraction, consider the mass of a chemical sample obtained by subtracting the mass of an empty weighing bottle (m_1) from the mass of the bottle filled with sample (m_2).

The mass of the empty bottle is measured as:

$$m_1 = 14.2315 \text{ g}$$

The absolute error in m_1 is:

$$s_{m_1} = 0.0002 \text{ g}$$

The mass of the filled bottle is measured as:

$$m_2 = 27.6842 \text{ g}$$

The absolute error in m_2 is:

$$s_{m_2} = 0.0003 \text{ g}$$

The mass of the sample m is calculated by subtracting m_1 from m_2:

$$m = m_2 - m_1$$

Substituting the measured masses gives the calculated sample mass:

$$m = 27.6842 \text{ g} - 14.2315 \text{ g} = 13.4527 \text{ g}$$

The absolute error in the sample mass s_m is calculated by adding the individual absolute errors in quadrature:

$$s_m = \sqrt{s_{m_1}^2 + s_{m_2}^2}$$

Substituting the absolute errors into the quadrature expression yields:

$$s_m = \sqrt{(0.0002\ \text{g})^2 + (0.0003\ \text{g})^2}$$

Evaluating the squared terms gives:

$$s_m = \sqrt{4.0 \times 10^{-8} + 9.0 \times 10^{-8}}\ \text{g}$$

Summing the terms inside the square root results in:

$$s_m = \sqrt{1.3 \times 10^{-7}}\ \text{g}$$

Taking the square root gives the final absolute error of the net sample mass:

$$s_m = 0.00036\ \text{g}$$

Thus, the final mass of the sample is reported as 13.4527 ± 0.0004 g.

Multiplication and Division

When measured quantities containing random uncertainties are multiplied or divided, the relative standard deviation of the final result is obtained by combining the relative standard deviations of individual terms in quadrature.

Consider a calculated quantity y derived from multiplication and division of measured quantities x_1, x_2, and x_3 :

$$y = \frac{x_1 \cdot x_2}{x_3}$$

The relative standard deviation of y is given by:

$$\frac{s_y}{y} = \sqrt{\left(\frac{s_{x_1}}{x_1}\right)^2 + \left(\frac{s_{x_2}}{x_2}\right)^2 + \left(\frac{s_{x_3}}{x_3}\right)^2}$$

To demonstrate error propagation in multiplication and division, consider the calculation of solute concentration C in mol L^{-1} from the measured number of moles n and solution volume V:

$$C = \frac{n}{V}$$

The measured number of moles is:

$$n = 0.0250\ \text{mol}$$

The absolute error in n is:

$$s_n = 0.0001\ \text{mol}$$

The measured volume of the solution is:

$$V = 0.5000\ \text{L}$$

The absolute error in V is:

$$s_V = 0.0010\ \text{L}$$

The molar concentration C is calculated as:

$$C = \frac{0.0250\ \text{mol}}{0.5000\ \text{L}} = 0.0500\ \text{mol L}^{-1}$$

The relative error of the number of moles is computed as:

$$\frac{s_n}{n} = \frac{0.0001}{0.0250} = 0.0040$$

The relative error of the volume is computed as:

$$\frac{s_V}{V} = \frac{0.0010}{0.5000} = 0.0020$$

The relative error of the calculated concentration C is given by quadrature:

$$\frac{s_C}{C} = \sqrt{\left(\frac{s_n}{n}\right)^2 + \left(\frac{s_V}{V}\right)^2}$$

Substituting the relative errors yields:

$$\frac{s_C}{C} = \sqrt{(0.0040)^2 + (0.0020)^2}$$

Evaluating the squared relative errors gives:

$$\frac{s_C}{C} = \sqrt{1.6 \times 10^{-5} + 4.0 \times 10^{-6}}$$

Summing the terms inside the square root gives:

$$\frac{s_C}{C} = \sqrt{2.0 \times 10^{-5}} = 0.00447$$

The absolute standard deviation s_C is determined by multiplying the relative error by the concentration C:

$$s_C = C \times \left(\frac{s_C}{C}\right)$$

Substituting the values yields:

$$s_C = 0.0500\ \text{mol L}^{-1} \times 0.00447 = 0.0002235\ \text{mol L}^{-1}$$

Rounding to one significant figure gives the final absolute error $s_C = 0.0002\ \text{mol L}^{-1}$. Thus, the concentration is reported as $0.0500 \pm 0.0002\ \text{mol L}^{-1}$.

4. Mean and Standard Deviation

Statistical measures provide quantitative methods for describing the central tendency and dispersion of replicate analytical measurement data.

Statistical Definitions and Mathematical Formulations

Arithmetic Mean

In quantitative chemistry, replicate measurements of a quantity are taken. For a set of n replicate measurements $x_1, x_2, \ldots, x_n$, the arithmetic mean $\bar{x}$ represents the central value of the data set.

The arithmetic mean is defined by:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Sample Standard Deviation

The sample standard deviation s measures the dispersion or spread of individual experimental data points around the sample mean. For a finite sample size n, s is calculated using $n - 1$ degrees of freedom in the denominator:

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}}$$

Sample Variance

The sample variance is defined as the square of the sample standard deviation, denoted as s^2:

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}$$

Relative Standard Deviation and Coefficient of Variation

To express precision independently of measurement units, the relative standard deviation (RSD) is defined as the ratio of the sample standard deviation to the arithmetic mean:

$$\text{RSD} = \frac{s}{\bar{x}}$$

When the relative standard deviation is expressed as a percentage, it is termed the coefficient of variation (%CV):

$$\%\text{CV} = \left(\frac{s}{\bar{x}} \right) \times 100$$

Standard Error of the Mean

The standard error of the mean $s_{\bar{x}}$ quantifies the uncertainty associated with using the sample mean $\bar{x}$ as an estimate of the true population mean. It is related to the sample standard deviation s and the sample size n by:

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}$$

Numerical Example: Evaluation of Replicate Titration Readings

Consider a series of five replicate burette volume readings obtained during the standardization of a sodium hydroxide solution against a primary standard acid:

$$x_1 = 10.12\ \text{mL}, \quad x_2 = 10.15\ \text{mL}, \quad x_3 = 10.11\ \text{mL}, \quad x_4 = 10.14\ \text{mL}, \quad x_5 = 10.13\ \text{mL}$$

The number of replicate readings is:

$$n = 5$$

The sum of the measured values is calculated as:

$$\sum_{i=1}^{5} x_i = 10.12 + 10.15 + 10.11 + 10.14 + 10.13 = 50.65\ \text{mL}$$

The arithmetic mean $\bar{x}$ is calculated by dividing the sum by the number of measurements:

$$\bar{x} = \frac{50.65\ \text{mL}}{5} = 10.130\ \text{mL}$$

The deviation $(x_i - \bar{x})$ for each reading from the mean is evaluated:

$$(x_1 - \bar{x}) = 10.12 - 10.130 = -0.010\ \text{mL}$$

$$(x_2 - \bar{x}) = 10.15 - 10.130 = +0.020\ \text{mL}$$

$$(x_3 - \bar{x}) = 10.11 - 10.130 = -0.020\ \text{mL}$$

$$(x_4 - \bar{x}) = 10.14 - 10.130 = +0.010\ \text{mL}$$

$$(x_5 - \bar{x}) = 10.13 - 10.130 = 0.000\ \text{mL}$$

The square of each individual deviation $(x_i - \bar{x})^2$ is computed:

$$(x_1 - \bar{x})^2 = (-0.010\ \text{mL})^2 = 0.000100\ \text{mL}^2$$

$$(x_2 - \bar{x})^2 = (+0.020\ \text{mL})^2 = 0.000400\ \text{mL}^2$$

$$(x_3 - \bar{x})^2 = (-0.020\ \text{mL})^2 = 0.000400\ \text{mL}^2$$

$$(x_4 - \bar{x})^2 = (+0.010\ \text{mL})^2 = 0.000100\ \text{mL}^2$$

$$(x_5 - \bar{x})^2 = (0.000\ \text{mL})^2 = 0.000000\ \text{mL}^2$$

The sum of the squared deviations is obtained by adding these values:

$$\sum_{i=1}^{5} (x_i - \bar{x})^2 = 0.000100 + 0.000400 + 0.000400 + 0.000100 + 0.000000 = 0.001000\ \text{mL}^2$$

The sample variance s^2 is calculated using $n - 1 = 4$ degrees of freedom:

$$s^2 = \frac{0.001000\ \text{mL}^2}{5-1} = \frac{0.001000}{4} = 0.000250\ \text{mL}^2$$

The sample standard deviation s is obtained by taking the square root of the variance:

$$s = \sqrt{0.000250\ \text{mL}^2} = 0.01581\ \text{mL}$$

The relative standard deviation (RSD) is calculated as:

$$\text{RSD} = \frac{0.01581\ \text{mL}}{10.130\ \text{mL}} = 0.00156$$

The percent coefficient of variation (%CV) is calculated as:

$$\%\text{CV} = 0.00156 \times 100 = 0.156\%$$

The standard error of the mean $s_{\bar{x}}$ is calculated as:

$$s_{\bar{x}} = \frac{0.01581\ \text{mL}}{\sqrt{5}} = \frac{0.01581}{2.23607} = 0.00707\ \text{mL}$$

Thus, the titration volume is reported as $10.130 \pm 0.016\ \text{mL}$ for individual measurements, or $\bar{x} \pm s_{\bar{x}} = 10.130 \pm 0.007\ \text{mL}$ for the estimated mean volume.

Applications of Mean and Standard Deviation in Chemical Analysis

The mean and standard deviation serve critical functions in analytical chemistry:

1. **Best Estimate of True Value:** In the absence of systematic error, the arithmetic mean of replicate measurements provides the most statistically reliable estimate of the population mean or true concentration of an analyte.
2. **Quantification of Precision:** The standard deviation provides a direct quantitative measure of experimental precision, repeatability, and analytical consistency. A smaller standard deviation corresponds to higher precision.
3. **Statistical Significance Testing and Confidence Intervals:** The mean and standard deviation are required to construct confidence intervals around an analytical result and to perform statistical tests (such as Student's t-test) to compare experimental means against target standard values or alternative analytical methods.
4. **Outlier Detection:** They are used in statistical tests (such as the Q-test or Dixon test) to objectively identify and reject suspect data points (outliers) within replicate data sets.

5. Linear Regression, Covariance and Correlation Coefficient

In quantitative chemical analysis, experimental measurement techniques frequently rely on establishing a functional relationship between a known physical quantity (such as analyte concentration, x) and a measured instrumental response (such as absorbance, potential, or peak area, y). To determine unknown sample

concentrations reliably, analytical chemists must evaluate how these two variables vary together, fit a mathematical model to the experimental data points, and evaluate the goodness of fit.

Covariance and Correlation Coefficient

Before fitting a line to experimental data, it is necessary to measure the direction and strength of the mutual relationship between two variables x and y. The sample covariance, denoted as $\operatorname{cov}(x, y)$, quantifies the joint variability of two random variables. For n pairs of measurements $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, the sample covariance is defined as:

$$\operatorname{cov}(x, y) = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

where $\bar{x}$ and $\bar{y}$ represent the arithmetic sample means of the independent variable x and dependent variable y, respectively. The sample means are defined as:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i$$

If x_i and y_i tend to deviate from their respective means in the same direction simultaneously (that is, when $x_i > \bar{x}$, $y_i > \bar{y}$ as well), the product $(x_i - \bar{x})(y_i - \bar{y})$ is positive, yielding a positive covariance. Conversely, if y_i decreases as x_i increases, the product is negative, yielding a negative covariance. If there is no systemic linear pattern, positive and negative terms cancel each other out, resulting in a covariance near zero.

Because the magnitude of $\operatorname{cov}(x, y)$ depends directly on the measurement units of x and y, it cannot be used directly as a normalized measure of association. To eliminate unit dependence, the sample covariance is divided by the product of the sample standard deviations of x and y. This defines the Pearson correlation coefficient, r:

$$r = \frac{\operatorname{cov}(x, y)}{s_x s_y}$$

The sample standard deviations s_x and s_y are defined by:

$$s_x = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}}$$

$$s_y = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \bar{y})^2}{n - 1}}$$

Substituting the explicit definitions of $\operatorname{cov}(x, y)$, s_x, and s_y into the expression for r yields:

$$r = \frac{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Canceling the common scale factor $\frac{1}{n-1}$ gives the standard computational formula for r:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

The correlation coefficient r is a dimensionless quantity restricted to the range:

$$-1 \leq r \leq +1$$

The value of r provides specific diagnostic information regarding the dataset: * $r \approx +1$: Indicates a strong positive linear correlation. An increase in x corresponds to a proportional linear increase in y. A value of $r = +1$ represents a perfect straight line with a positive slope. * $r \approx 0$: Indicates the absence of any linear relationship between x and y. Changes in x provide no linear information about y. * $r \approx -1$: Indicates a strong negative linear correlation. An increase in x corresponds to a proportional linear decrease in y. A value of $r = -1$ represents a perfect straight line with a negative slope.

Linear Regression by the Method of Least Squares

When a linear relationship between analyte concentration x and instrumental response y is assumed, the ideal response model is represented by the equation of a straight line:

$$y = mx + c$$

where m is the slope (representing detector sensitivity) and c is the y-intercept (representing the background signal or blank response).

Due to random experimental errors, individual measured data points (x_i, y_i) do not fall perfectly on an ideal straight line. For a given value x_i, the model predicts a response $\hat{y}_i$:

$$\hat{y}_i = mx_i + c$$

The residual e_i is defined as the vertical difference between the actual observed experimental response y_i and the model-predicted response $\hat{y}_i$:

$$e_i = y_i - \hat{y}_i$$

Substituting the linear model into the residual definition gives:

$$e_i = y_i - (mx_i + c)$$

The method of ordinary least squares assumes that errors in the independent calibration variable x are negligible compared to errors in the measured dependent response y. The best-fit straight line is defined as the unique line that minimizes the sum of the squared residuals, denoted by S:

$$S = \sum_{i=1}^{n} e_i^2$$

Substituting the explicit expression for e_i into S gives:

$$S = \sum_{i=1}^{n} (y_i - mx_i - c)^2$$

To minimize S with respect to the parameters c and m, we calculate the partial derivatives of S with respect to c and m independently, setting each derivative equal to zero.

Taking the partial derivative of S with respect to the intercept c gives:

$$\frac{\partial S}{\partial c} = -2 \sum_{i=1}^{n} (y_i - mx_i - c) = 0$$

Dividing both sides of the equation by -2 yields:

$$\sum_{i=1}^{n} (y_i - mx_i - c) = 0$$

Distributing the summation operator across each term gives:

$$\sum_{i=1}^{n} y_i - m \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} c = 0$$

Recognizing that summing a constant c over n points equals nc, we write:

$$\sum_{i=1}^{n} y_i - m \sum_{i=1}^{n} x_i - nc = 0$$

Rearranging the terms to isolate nc produces:

$$nc = \sum_{i=1}^{n} y_i - m \sum_{i=1}^{n} x_i$$

Dividing both sides by n gives:

$$c = \frac{1}{n} \sum_{i=1}^{n} y_i - m \left(\frac{1}{n} \sum_{i=1}^{n} x_i \right)$$

Substituting the sample means $\bar{y} = \frac{1}{n} \sum y_i$ and $\bar{x} = \frac{1}{n} \sum x_i$ yields the explicit formula for the intercept c:

$$c = \bar{y} - m\bar{x}$$

Next, taking the partial derivative of S with respect to the slope m gives:

$$\frac{\partial S}{\partial m} = -2\sum_{i=1}^{n} x_i(y_i - mx_i - c) = 0$$

Dividing both sides of this equation by -2 yields:

$$\sum_{i=1}^{n} x_i(y_i - mx_i - c) = 0$$

Substituting the derived relation $c = \bar{y} - m\bar{x}$ into the equation gives:

$$\sum_{i=1}^{n} x_i(y_i - mx_i - (\bar{y} - m\bar{x})) = 0$$

Regrouping the terms within the parentheses yields:

$$\sum_{i=1}^{n} x_i\left((y_i - \bar{y}) - m(x_i - \bar{x})\right) = 0$$

Expanding the summation across the terms yields:

$$\sum_{i=1}^{n} x_i(y_i - \bar{y}) - m\sum_{i=1}^{n} x_i(x_i - \bar{x}) = 0$$

Using the mathematical identities $\sum x_i(y_i - \bar{y}) = \sum(x_i - \bar{x})(y_i - \bar{y})$ and $\sum x_i(x_i - \bar{x}) = \sum(x_i - \bar{x})^2$, we rewrite the equation as:

$$\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) - m\sum_{i=1}^{n}(x_i - \bar{x})^2 = 0$$

Isolating the term containing m produces:

$$m\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Solving directly for the slope m gives the least-squares formula:

$$m = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

An equivalent computational form expressed using raw sums is:

$$m = \frac{n\sum_{i=1}^{n} x_i y_i - \left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2}$$

Once the optimal slope m and intercept c are determined, the precision of the fit is quantified using the standard error of the estimate, denoted as $s_{y/x}$. This statistic measures the dispersion of the observed y-values around the

regression line:

$$s_{y/x} = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n - 2}}$$

Substituting the regression model $\hat{y}_i = m x_i + c$ into the equation yields:

$$s_{y/x} = \sqrt{\frac{\sum_{i=1}^{n} (y_i - m x_i - c)^2}{n - 2}}$$

The denominator contains $n - 2$ degrees of freedom because two independent parameters (the slope m and intercept c) have been evaluated from the dataset. A smaller value of $s_{y/x}$ indicates that experimental points lie close to the calculated calibration curve.

Use of Regression and Correlation in Chemical Analysis

Linear regression and correlation analysis serve critical functions in standard analytical methodology:

1. **Calibration Curves**: In analytical techniques such as spectrophotometry, atomic absorption, and chromatography, regression analysis is used to convert instrumental responses into calibration functions. Once m and c are established, the unknown concentration x_{unk} of a test sample is calculated from its measured signal y_{unk} using the rearranged equation:

$$x_{\text{unk}} = \frac{y_{\text{unk}} - c}{m}$$

1. **Checking Linearity**: The correlation coefficient r (and its square, the coefficient of determination r^2) is evaluated to check whether the detector response remains strictly linear over the chosen concentration range. In routine quantitative analysis, an absolute correlation value of $|r| \geq 0.995$ or $r^2 \geq 0.99$ is typically required to confirm linear dynamic range.
2. **Evaluating Blank Corrections**: The calculated intercept c provides an empirical estimate of the reagent blank signal, allowing analysts to determine whether significant background interference is present.

Worked Numerical Example

Consider a spectrophotometric experiment where standard solutions of a chemical species with known concentrations x_i (in mg L^{-1}) give the measured absorbance values y_i shown below for $n = 5$ calibration points.

Point (i)	Concentration x_i (mg L^{-1})	Absorbance y_i
1	1.00	0.150
2	2.00	0.280
3	3.00	0.420
4	4.00	0.550

Point (i)	Concentration x_i ($\mathrm{mg\ L^{-1}}$)	Absorbance y_i
5	5.00	0.700

To compute the slope m, intercept c, covariance $\mathrm{cov}(x,y)$, and correlation coefficient r, we first calculate the necessary sample statistics.

The sample mean of x is:

$$\bar{x} = \frac{1.00 + 2.00 + 3.00 + 4.00 + 5.00}{5} = \frac{15.00}{5} = 3.00\ \mathrm{mg\ L^{-1}}$$

The sample mean of y is:

$$\bar{y} = \frac{0.150 + 0.280 + 0.420 + 0.550 + 0.700}{5} = \frac{2.100}{5} = 0.420$$

Next, we evaluate the deviation terms $(x_i - \bar{x})$, $(y_i - \bar{y})$, their squares, and their cross-products:

i	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
1	-2.00	-0.270	4.0000	0.0729	0.5400
2	-1.00	-0.140	1.0000	0.0196	0.1400
3	0.00	0.000	0.0000	0.0000	0.0000
4	1.00	0.130	1.0000	0.0169	0.1300
5	2.00	0.280	4.0000	0.0784	0.5600
$\sum$	0.00	0.000	10.0000	0.1878	1.3700

Using the calculated sums, we evaluate the slope m:

$$m = \frac{\sum_{i=1}^{5}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{5}(x_i - \bar{x})^2}$$

Substituting the tabular sums into the slope formula gives:

$$m = \frac{1.3700}{10.0000} = 0.1370\ \mathrm{L\ mg^{-1}}$$

Using the calculated value of m, the intercept c is:

$$c = \bar{y} - m\bar{x}$$

Substituting $\bar{y} = 0.420$, $m = 0.1370$, and $\bar{x} = 3.00$ yields:

$$c = 0.420 - (0.1370 \times 3.00) = 0.420 - 0.411 = 0.0090$$

The empirical calibration equation fitted to the analytical data is:

$$y = 0.1370x + 0.0090$$

Next, we evaluate the sample covariance $\operatorname{cov}(x, y)$:

$$\operatorname{cov}(x, y) = \frac{\sum_{i=1}^{5} (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Substituting $n = 5$ and the cross-product sum yields:

$$\operatorname{cov}(x, y) = \frac{1.3700}{5 - 1} = \frac{1.3700}{4} = 0.3425$$

The sample standard deviation of x is:

$$s_x = \sqrt{\frac{\sum_{i=1}^{5} (x_i - \bar{x})^2}{n - 1}} = \sqrt{\frac{10.0000}{4}} = \sqrt{2.5000} \approx 1.5811$$

The sample standard deviation of y is:

$$s_y = \sqrt{\frac{\sum_{i=1}^{5} (y_i - \bar{y})^2}{n - 1}} = \sqrt{\frac{0.1878}{4}} = \sqrt{0.04695} \approx 0.2167$$

The Pearson correlation coefficient r is evaluated using the formula:

$$r = \frac{\operatorname{cov}(x, y)}{s_x s_y}$$

Substituting the calculated values yields:

$$r = \frac{0.3425}{1.5811 \times 0.2167} = \frac{0.3425}{0.3426} \approx 0.9997$$

Alternatively, calculating r directly from the unscaled sum of squares gives:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Substituting the values yields:

$$r = \frac{1.3700}{\sqrt{10.0000 \times 0.1878}} = \frac{1.3700}{\sqrt{1.8780}} = \frac{1.3700}{1.3704} = 0.9997$$

The value $r = 0.9997$ lies extremely close to $+1$, confirming an excellent linear response over the evaluated concentration range.

6. Significant Figures

Concept and Importance of Significant Figures

In quantitative chemical analysis, every measurement possesses an inherent uncertainty determined by the limitations of the measuring instrument and the technique employed. The numerical value recorded from an experimental measurement consists of all the certain digits plus the first uncertain or estimated digit. These digits are collectively defined as significant figures.

Significant figures reflect the precision of an experimental measurement. The fundamental principle governing their use in analytical data treatment is that an analytical result cannot be more precise than the least precise measurement used to obtain that result. Retaining too many digits in a reported value implies a degree of experimental precision that does not exist, whereas discarding digits arbitrarily eliminates valid scientific information.

Rules for Determining Significant Figures

To establish the number of significant figures present in any measured quantity, five foundational rules are applied:

1. **All non-zero digits are significant.** Any non-zero digit appearing in a measurement represents a functional part of the measured quantity.
2. **Zeros situated between non-zero digits are significant.** These are often called captive or confined zeros.
3. **Leading zeros are not significant.** Zeros that precede the first non-zero digit serve only to locate the decimal point and specify the magnitude of the number; they do not reflect experimental precision.
4. **Trailing zeros after a decimal point are significant.** Zeros at the end of a number that contains a decimal point indicate the precision of the measuring instrument.
5. **Trailing zeros in a whole number without an explicit decimal point are ambiguous.** Such zeros may represent measured precise quantities or merely placeholders indicating magnitude. To remove this ambiguity, scientific notation must be used.

Worked Examples for Identification Rules

Applying the first rule, the mass measurement $14.372\ \text{g}$ contains five non-zero digits, so it has five significant figures:

$$\text{Number of significant figures in } 14.372\ \text{g} = 5$$

Applying the second rule, the volume reading $100.04\ \text{mL}$ contains three captive zeros between non-zero digits (1 and 4), all of which are significant:

$$\text{Number of significant figures in } 100.04\ \text{mL} = 5$$

Applying the third rule, the concentration value $0.00045\ \text{M}$ contains four leading zeros preceding the digit 4. These leading zeros are not significant, leaving only the digits 4 and 5:

$$\text{Number of significant figures in } 0.00045 \text{ M} = 2$$

Applying the fourth rule, a burette reading reported as $25.00\ \text{mL}$ includes two trailing zeros after the decimal point, demonstrating precision to the hundredths place:

$$\text{Number of significant figures in } 25.00 \text{ mL} = 4$$

Applying the fifth rule, the volume $500\ \text{mL}$ lacks an explicit decimal point and is ambiguous regarding whether it has one, two, or three significant figures. Expressing this quantity in scientific notation resolves all ambiguity explicitly:

$$\text{For one significant figure: } 5 \times 10^2 \text{ mL}$$

$$\text{For two significant figures: } 5.0 \times 10^2 \text{ mL}$$

$$\text{For three significant figures: } 5.00 \times 10^2 \text{ mL}$$

Rules for Rounding Numerical Values

When performing calculations, intermediate results often generate more digits than are significant. The calculated value must be rounded off to the correct number of significant figures using standard analytical rounding rules:

1. **If the digit to be dropped is less than** 5, the preceding digit remains unchanged (round down).
2. **If the digit to be dropped is** 5 **or more**, the preceding digit is increased by 1 (round up).
3. **For the specific convention where the digit to be dropped is exactly** 5 **followed by nothing (or followed only by zeros)**, the standard round-to-even rule is frequently applied in analytical chemistry: if the preceding digit is even, it remains unchanged; if the preceding digit is odd, it is increased by 1 to make it even. Alternatively, under general standard rounding rules, any value of 5 or greater causes the preceding digit to round up.

Worked Examples for Rounding Rules

Rounding the number $6.734\ \text{g}$ to three significant figures requires dropping the digit 4. Because 4 is less than 5, the preceding digit 3 is left unchanged:

$$6.734\ \text{g} \longrightarrow 6.73\ \text{g}$$

Rounding the number $6.738\ \text{g}$ to three significant figures requires dropping the digit 8. Because 8 is greater than 5, the preceding digit 3 is increased by 1:

$$6.738\ \text{g} \longrightarrow 6.74\ \text{g}$$

Rounding the number $6.735\ \text{g}$ to three significant figures involves dropping an exact 5 followed by nothing. Under the round-to-even convention, because the preceding digit 3 is odd, it is increased by 1 to become even:

$$6.735\ \text{g} \longrightarrow 6.74\ \text{g}$$

Rounding the number $6.745\ \text{g}$ to three significant figures under the same round-to-even convention leaves the digit 4 unchanged because 4 is already even:

$$6.745\ \text{g} \longrightarrow 6.74\ \text{g}$$

Significant Figures in Calculations

Addition and Subtraction

In addition and subtraction operations, the precision of the final answer is limited by the measurement with the fewest decimal places. The absolute magnitude of the number of significant figures does not dictate the rounding; rather, the position of the decimal point determines the result.

The mathematical sum of three analytical masses m_1, m_2, and m_3 is expressed as:

$$m_{\text{total}} = m_1 + m_2 + m_3$$

Substituting the measured masses $m_1 = 12.11\ \text{g}$ (two decimal places), $m_2 = 18.0\ \text{g}$ (one decimal place), and $m_3 = 1.013\ \text{g}$ (three decimal places) into the summation yields:

$$m_{\text{total}} = 12.11\ \text{g} + 18.0\ \text{g} + 1.013\ \text{g}$$

Performing the unrounded addition gives:

$$m_{\text{total}} = 31.123\ \text{g}$$

Because $m_2 = 18.0\ \text{g}$ has the fewest decimal places (one decimal place), the final result must be rounded to one decimal place. Dropping the digits $.023$ (where the first dropped digit is $2 < 5$) gives the correctly formatted reported mass:

$$m_{\text{total}} = 31.1\ \text{g}$$

Consider a subtraction operation calculating mass difference Δm:

$$\Delta m = m_{\text{final}} - m_{\text{initial}}$$

Substituting the values $m_{\text{final}} = 24.582\ \text{g}$ (three decimal places) and $m_{\text{initial}} = 2.1\ \text{g}$ (one decimal place):

$$\Delta m = 24.582\ \text{g} - 2.1\ \text{g}$$

The unrounded difference is:

$$\Delta m = 22.482\ \text{g}$$

Rounding to one decimal place (since m_{initial} has only one decimal place) and observing that the first dropped digit is $8 \geq 5$, the final value rounds up:

$$\Delta m = 22.5\ \text{g}$$

Multiplication and Division

In multiplication and division operations, the final result must contain the same number of significant figures as the measurement containing the fewest significant figures.

The mass density ρ of a substance is defined as the ratio of mass m to volume V:

$$\rho = \frac{m}{V}$$

Substituting an experimental mass $m = 4.237\ \text{g}$ (four significant figures) and an experimental volume $V = 1.2\ \text{mL}$ (two significant figures) into the density formula gives:

$$\rho = \frac{4.237\ \text{g}}{1.2\ \text{mL}}$$

Evaluating the quotient yields the unrounded value:

$$\rho = 3.530833...\ \text{g/mL}$$

Because the volume measurement $1.2\ \text{mL}$ contains the fewest significant figures (two significant figures), the density must be reported to two significant figures. Dropping the trailing digits $.030833...$ (where the first dropped digit is $3 < 5$) yields:

$$\rho = 3.5\ \text{g/mL}$$

Consider another example computing the number of moles n of a solute:

$$n = C \times V$$

Substituting a concentration $C = 0.154\ \text{M}$ (three significant figures) and a solution volume $V = 0.0250\ \text{L}$ (three significant figures) into the equation yields:

$$n = 0.154\ \text{mol/L} \times 0.0250\ \text{L}$$

Evaluating the product gives:

$$n = 0.00385\ \text{mol}$$

Both measurements contain three significant figures (note that leading zeros in 0.154 and 0.0250 are not significant, while the trailing zero in 0.0250 is significant). The answer already contains exactly three significant figures (3, 8, and 5), so no rounding is necessary:

$$n = 3.85 \times 10^{-3}\ \text{mol}$$

Questions

Short form first (2–3 marks), then long form (5–10 marks). **There are 8 Assam University questions for this unit in the paper bank**, from CHMDSE-501T Analytic Methods in Chemistry, 5th Sem, 2020 — **not from a DSM-401 paper**; they match Unit I topics (sampling, errors, precision/accuracy, data analysis) only. The other questions are from other sources, marked as such.

Short-form questions

S1. [AU] What do you mean by the term 'sampling'? — CHMDSE-501T Analytic Methods in Chemistry, 5th Sem, 2020, CBCS (Sec A, 1 mark).

Answer in brief: Section 1 — taking a small representative portion of the bulk material for analysis; the sample must represent the whole.

S2. [AU] Define error. — CHMDSE-501T, 2020, CBCS (Sec A, 1 mark).

Answer in brief: Section 2 — the difference between a measured value and the true value.

S3. [AU] What is meant by precision? — CHMDSE-501T, 2020, CBCS (Sec A, 1 mark).

Answer in brief: Section 2 — the reproducibility of repeated measurements of the same quantity.

S4. [AU] Define accuracy. — CHMDSE-501T, 2020, CBCS (Sec A, 1 mark).

Answer in brief: Section 2 — the closeness of a measured value to the true value.

S5. [AU] What are different ways of expressing an error? — CHMDSE-501T, 2020, CBCS (Sec A, 1 mark).

Answer in brief: Section 2 — absolute error, relative error, percentage error, parts per thousand.

S6. [standard] What are significant figures? State the rules with examples.

Answer in brief: Section 6 — the meaningful digits in a measurement; non-zero digits and zeros between them count, leading zeros do not, trailing zeros after a decimal point count.

Long-form questions

L1. [AU] Discuss how you can minimize error. — CHMDSE-501T, 2020, CBCS (Sec B, 2 marks).

What to write: Sections 2–3 in full — calibration, blanks, replicate measurements, standard reference materials, and removing systematic errors at their source.

L2. [AU] What is meant by normal distribution? Give examples. What are the characteristics of normal distribution? — CHMDSE-501T, 2020, CBCS (Sec C, 5 marks).

What to write: Section 1 in full — the bell-shaped distribution of random errors about the mean, the 68-95-99.7 rule, and examples such as repeated titration readings.

L3. [AU] Discuss different statistical tests of data analysis. — CHMDSE-501T, 2020, CBCS (Sec C, 5 marks).

What to write: Section 1 in full — the Q-test for rejecting an outlier, the t-test for comparing means, and the F-test for comparing precisions.

L4. [standard] Define sampling and explain evaluation of analytical data.

What to write: Section 1 in full — types of sampling, representativeness, the normal distribution, and the Q-, t- and F-tests.

L5. [standard] Distinguish between precision and accuracy with examples; define mean and standard deviation and state their use in chemical analysis.

What to write: Sections 2 and 4 in full — the target-and-arrow comparison, mean $\bar{x} = (\sum x_i)/n$, standard deviation s, and the worked titration example.

L6. [standard] Explain systematic errors and random errors with examples, and explain propagation of errors.

What to write: Section 3 in full — determinate vs indeterminate errors with examples, and the quadrature rules for addition/subtraction and multiplication/division with the worked examples.

L7. [standard] Define linear regression, covariance and correlation coefficient and explain their use.

What to write: Section 5 in full — least-squares line $y = mx + c$, covariance, $r = \mathrm{cov}(x, y)/(s_x s_y)$ with its range and meaning, and the calibration-curve example.

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.